How Do the Volatilities of Rates Depend on Their Level?

The "Universal Relationship" Revisited

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KEY FINDINGS

- The universal relationship, across currencies and tenors, between volatilities and the level of rates first presented in De Guillaume, Rebonato, and Pogudin (2013) also applies to the extremely-low-rate environment of the last decade, and we present a simple extension of the relationship to handle the case of negative rates.
- We explain the origin of this relationship by showing the existence of two sharply distinct regimes for the volatility of real rates as a function of real rate levels, and by linking periods of high inflation with periods of high real rates.
- We show that the "volatility of volatility" also displays a "universal" behaviour, with a significant linear dependence on the level of rates (and of the volatility itself).

ABSTRACT

We present a straightforward extension valid in the current negative-rate regime of the "universal relationship" uncovered in De Guillaume, Rebonato, and Pogudin (2013) between the level of rates and their volatility. We also provide an explanation of the origin of this relationship by showing the existence of two sharply distinct regimes for the volatility of real rates as a function of real rate levels, and by linking periods of high inflation with periods of high real rates. Finally, we provide evidence that the "volatility of volatility" also displays a "universal" behavior, with a significant linear dependence on the level of rates (and of the volatility itself).

TOPICS

Fixed income and structured finance, volatility measures, quantitative methods, statistical methods, developed markets*

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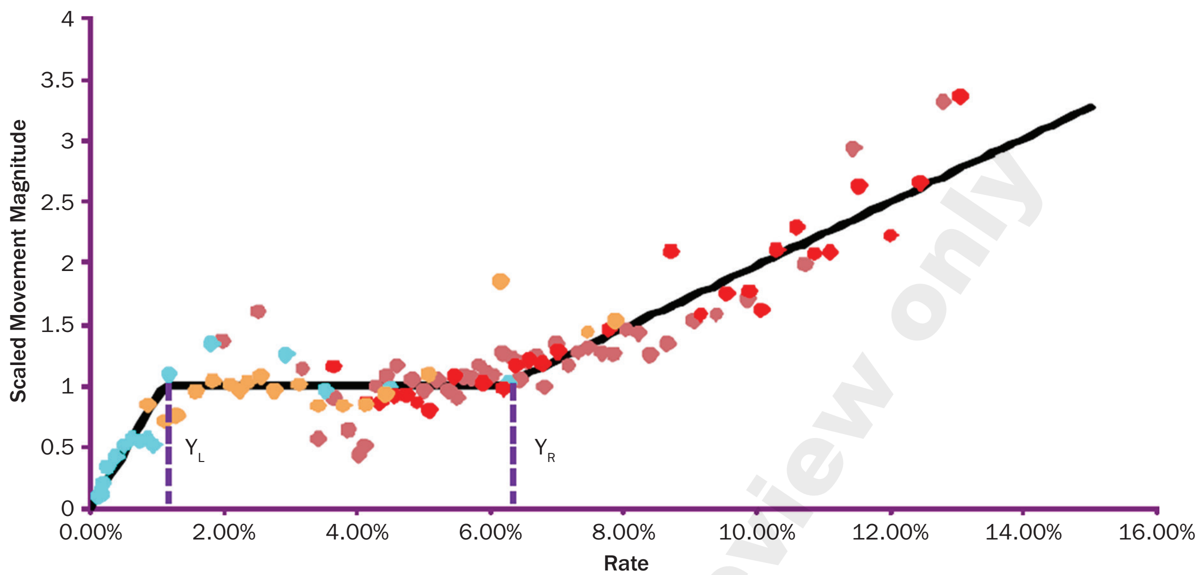
In their 2013 paper De Guillaume, Rebonato, and Pogudin (2013) highlighted an unexpected and, in their words, "universal" relationship between the volatility of rates and their level. In a nutshell, they showed that, irrespective of country or tenor, all risk-less rates display a three-regime behavior: when rates are very low (say, below 2%), their volatility is approximately proportional to their level; between 2% and approximately 6% there is no dependence of their volatility on the rate level; and above 6% rate changes are again proportional in magnitude to the rate level. Mindful that one well-chosen figure can really be worth a thousand words, we reproduce in Exhibit 1 their key graphical result, which conveys the essence of their story more eloquently and succinctly than we could.

EXHIBIT 1

Rescaled Magnitude of the Movement in Rates as a Function of the Rate Level (in percentage points)



De Guillaume, Rebonato, and Pogudin buttressed their universality claim by 1) introducing a clever standardization for the volatility (under which the level dependence becomes very apparent); 2) jointly analyzing risk-less rates in US dollars (USD), Swiss Francs (CHF), British pounds (GBP), Japanese yens (JPN), and euros (EUR); and 3) going back in time all the way to the 19th century (with the GBP consol yields). Most surprisingly, they showed that, after an appropriate rescaling, the "switch points" between regimes and the parameters characterizing each regime are only very mildly dependent on the currency, on the tenor or the exact nature of the rates (yields to maturity, swap rates, LIBOR rates, etc.); the switch points and the remaining parameters of their model are, in other words, close to universal.

Finding such a universal relationship is clearly important, as it shows that at least part of the variability of rates is of the "local-volatility" type, and this dependence is the same across tenors and currencies. This finding therefore suggests that a useful diffusive description of the behavior of rates can be accomplished along the following lines:

$$dr(t) = \mu dt + \underbrace{s(r_t)g(t)}_{\sigma(r,t)} dz_t \tag{1}$$

$$dg(t) = m dt + \nu dw_t \tag{2}$$

where μ and m denote the real-world or risk-neutral drifts for the rate, $r(t)$, and for its volatility, respectively, and with the term ν (the "volatility of volatility") capturing the residual degree of variability in rates not captured by their level. The universal relationship uncovered by De Guillaume, Rebonato, and Pogudin then pins down the function $s(r_t)$ for rates of any maturity and currency.

Uncovering the existence and nature of the level dependence of volatility on the level of rates is also important for resampling techniques, such as the recently introduced Crump and Gospodinov (2019) bootstrapping method. In all these applications, sections of past yield histories are sequentially added to an initial yield curve segment. As the synthetic yield curve is created, its level changes over time. If the level of synthetic rates thus generated were, say, very low, and the volatility of rates depended in a predictable way on their level, it would be inappropriate to “glue” in this low-rate environment a new segment drawn from a high-rate regime, and vice versa. Given the current near-ubiquity of resampling techniques for non-parametric studies of statistical significance (see, e.g., Clark and McCracken 2013 for a recent review), the existence of a dependence between rate variability and level poses an important, if poorly appreciated, problem.

De Guillaume, Rebonato, and Pogudin (2013) solve this problem by creating a “disguise” for the original yields—that is, a transformation under which the level dependence disappears. Once so disguised, yield changes can safely be sampled and added to an existing synthetic (disguised) curve, without having to worry about where the yields changes came from. Once the bootstrapping in the disguised-yield space is carried out, the rates can be “undisguised” and turned into their natural units.[1]

The present work extends the results in De Guillaume, Rebonato, and Pogudin (2013) in three directions. First, when the authors presented their results, negative rates for government bonds had never been observed; now they can be found in all the currencies under study with the exception (so far) of USD. As a consequence, a log-normal behavior cannot be appropriate for very low levels of rates. We therefore modify the functional form of the dependence of volatility on rates for low levels of rates to accommodate the case of negative rates.

The second contribution of the present work is arguably more interesting. The relationship De Guillaume, Rebonato, and Pogudin presented may well have been universal, but it also was to some extent magic, as no explanation was offered for the regularities they discovered. In this paper, we fall short of providing a full explanation, but we at least push the magic element a bit farther down the line. We show in fact that the three-regime behavior uncovered in De Guillaume, Rebonato, and Pogudin (2013) comes from the interplay of very different behaviors for the volatility of real rates and inflation. Inflation volatility in fact shows very little dependence on its level; the volatility of real rates displays instead a more complex behavior, with two distinct regimes for high and low levels of real rates. As we shall show, the “upper branch” of this bifurcated pattern for real-rate volatilites is associated with conditions of high inflation. The combination of real rate volatility with inflation-level-independent volatility of inflation for high inflation then gives rise to the pattern observed in De Guillaume, Rebonato, and Pogudin (2013)—and confirmed in this study.

Third, we examine the “residual” of the variability of rates once the level dependence has been stripped out, and find that the remaining variability (the “volatility of volatility”) is directly proportional to the level of rates. And, since the highest variability is associated with the highest rates, this means that the more uncertain we are (the higher the level of rates), the greater our uncertainty about how uncertain we are. Since the vast majority of stochastic-volatility models of which we are aware posit a constant volatility of volatility, our finding can provide guidance for a simple modeling extension.

The paper is therefore organized as follows. To provide a self-contained presentation, the next section gives a concise account of the disguise procedure presented in De Guillaume, Rebonato, and Pogudin (2013) for the positive-rate case. The procedure is then extended to the negative-rate case in the following section. Then we present the data used for the empirical study, followed by the results, and then provide an explanation for the observed universal pattern. We draw our conclusions in the final section.

[1] A closely related application is in the historical-simulation method for Value-at-Risk calculations, as described, for instance, in Chapter 13 of Hull (2018).

THE DISGUISE METHOD: SIGMA FUNCTIONS AND THEIR INVERSE

For clarity of exposition we cast the procedure in terms of a re-sampling exercise, but, as discussed in the introduction, the approach lends itself to more general applications.

We denote "today" as time k and an arbitrary past date as ω_i. Call the rate today $y(k)$ and the rate observed in the past $Y(\omega_i)$. Suppose that we want to evolve today's rate by one time step in the objective measure, i.e., that we want to simulate $y(k+1)$. To create this change in today's rate (i.e., $y(k+1) - y(k)$) we want to use information about the change that occurred on some day in the past, $Y(\omega_{i+1}) - Y(\omega_i)$.

To do so, De Guillaume, Rebonato, and Pogudin introduce a class of functions they call σ functions, $\sigma: \mathbb{R}^+ \Rightarrow \mathbb{R}$. Members of this class of function can all be expressed in the form

$$\begin{aligned} &= \sigma_G \frac{y}{y_L} && 0 < y < y_L \\ \sigma(y) &= \sigma_G && y_L \le y < y_R \end{aligned} \tag{3}$$

$$= \sigma_G[1 + K(y - y_R)] \qquad y_R \le y \tag{4}$$

Given a σ function, its corresponding Σ function, $\Sigma: \mathbb{R}^+ \Rightarrow \mathbb{R}$, is then defined by

$$\Sigma(y) \equiv y_L + \sigma_G \int_{y_L}^{y} \frac{ds}{\sigma(s)} \tag{5}$$

The function $\Sigma(y)$ can be explicitly computed as

$$\begin{aligned} &= y_L\left(1 + \log\left(\frac{y}{y_L}\right)\right) && 0 < y < y_L \\ \Sigma(y) &= y && y_L \le y < y_R \\ &= y_R + \frac{1}{K}\log(1+(y - y_R)) && y_R \le y \end{aligned} \tag{6}$$

and its inverse, $\Sigma^{-1}(\nu)$, $\Sigma^{-1}: \mathbb{R} \Rightarrow \mathbb{R}^+$, is given by

$$\begin{aligned} &= \Sigma_L \exp\left(\frac{v}{\Sigma_L} - 1\right) && v \le \Sigma_L \\ \Sigma^{-1}(v) &= v && y_L \le y < y_R \\ &= \Sigma_R + \frac{1}{K}(\exp(K(v - \Sigma_R)) - 1) && \Sigma_R \le v \end{aligned} \tag{7}$$

with $\Sigma_L = \Sigma(y_L)$ and $\Sigma_R = \Sigma(y_R)$. The inverse, $\Sigma^{-1}(\nu)$, then takes one back to rates in the real-world measure. De Guillaume, Rebonato, and Pogudin call the application of the Σ function a "disguise" because, after the transformation, one should no longer be able to tell whether a segment of a bootstrapped yield curve "came from" a high- or a low-rate period. (See Figure 10 in De Guillaume, Rebonato, and Pogudin 2013, and Exhibits 2 and 3 in this paper.)

The sigma function is defined by the upper and lower switching points (y_L and y_R) and by the constant, K. To determine these, De Guillaume, Rebonato, and

EXHIBIT 2

Original and Disguised Yields for USD (left panel) and EUR (right panel) Obtained with the Modified Sigma Function

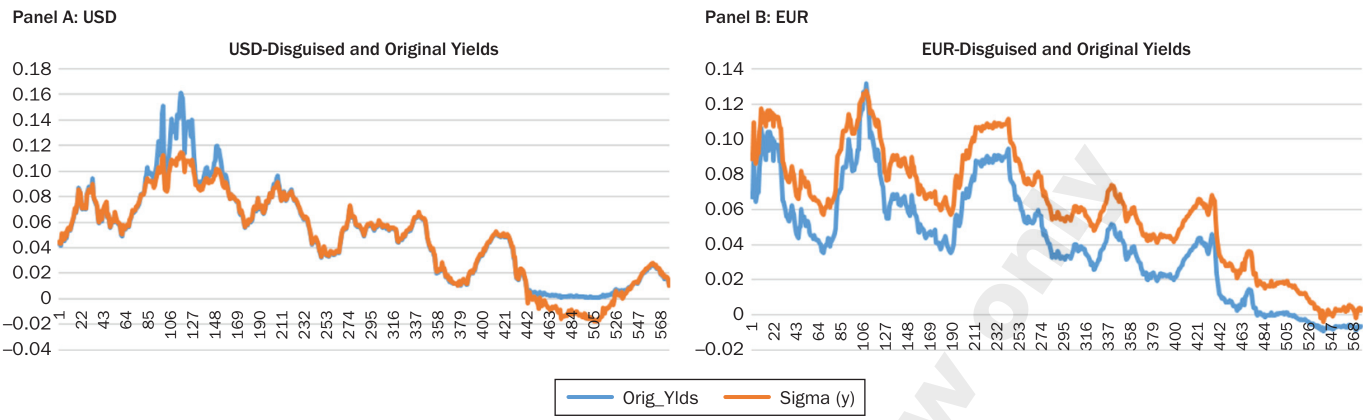
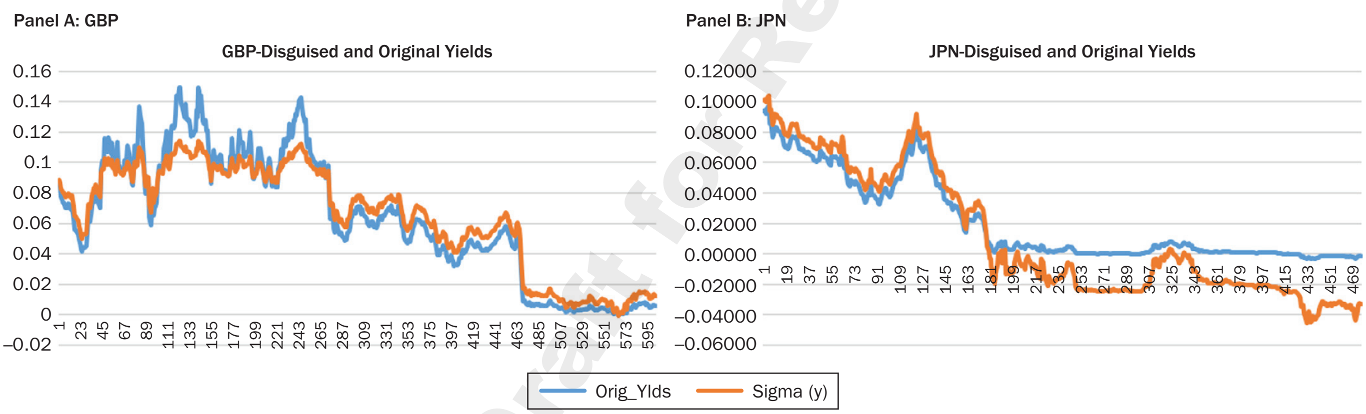


EXHIBIT 3

The Original and Disguised Yields for GBP (left panel) and JPN (right panel) Obtained with the Modified Sigma Function



Pogudin (2013) first define a candidate Σ function by choosing initial values for y_L, y_R, and K. After lightening notation by defining $\Sigma(y) = \bar{Y}$, they take the time-series $y(j)_{j \le m+s}$ and apply Σ to create first a time series $\bar{Y}(j)_{j \le m+s}$, and then a sequence of s-day absolute changes $(\Delta(j))_{j<m}$ from $\bar{Y}(j)_{j \le m+s}$ using the formula $\Delta(j) = \bar{Y}(j+s) - \bar{Y}(j)$. Let μ be the average of the Δ series and σ'_G its average deviation defined by the formula

$$\sigma'_G = \frac{1}{m}\sum_{j=1}^{m} |\Delta(j) - \mu| \tag{8}$$

The calibration metric is then given by

$$\frac{1}{\sigma'_G}\sum_{j=1}^{m} \| \Delta(j) - \mu | - \sigma'_G | \tag{9}$$

and assigns "penalties" depending on how different the absolute movements of the converted time series are from the average absolute movement. A search algorithm is then used to find the parameters y_L, y_R, and K that minimize the calibration metric.[2]

Exhibit 1 shows the clear universal pattern described in De Guillaume, Rebonato, and Pogudin (2013) for positive rates. The figure is produced as described in Appendix I.

Appendix I and De Guillaume, Rebonato, and Pogudin (2013) contain additional computational details, but, in a nutshell, the procedure used to obtain Exhibit 1 is a standardization mechanism: in some currencies rates may be "intrinsically" more volatile (as De Guillaume, Rebonato, and Pogudin say), but, after correcting for this currency-specific feature via the quantity σ'_G, one can analyze the volatilities of different currencies on the same footing. We therefore also employ this standardization device in the rest of the analysis.

THE MODIFIED SIGMA FUNCTION

It is clear that Equation 3 cannot hold in a negative-rate environment. We therefore take inspiration from the work by Rubinstein (1983), who introduces displaced diffusions by requiring that the logarithm of $(y + a)$, rather than the rate y itself, should be normally distributed:

$$\frac{d(r_t + a)}{r_t + a} = \frac{dr_t}{r + a} = \mu dt + \sigma dz \tag{10}$$

Given this specification, rates can attain all values greater than $-a$, usually referred to as the displacement coefficient. For the properties of displaced diffusions in the context of volatility modeling see Svoboda-Greenwood (2007), Marris (1999), Joshi and Rebonato (2003), and Lee and Wang (2009).

For our purposes, we assume a displaced-diffusion behavior for the lowest rates, and we impose the continuity across the three segments of the Sigma function and of its first derivative. When we do so, we obtain the following modified behavior for the Sigma Function:

$$\Sigma(y) \begin{array}{ll} = (y_L + \alpha)\left(1 + \log\left(\dfrac{y + \alpha}{y_L + \alpha}\right)\right) & 0 < y < y_L \\ = y + \alpha & y_L \le y < y_R \\ = y_R + \alpha + \dfrac{1}{K}\log(1 + (y - y_R)) & y_R \le y \end{array} \tag{11}$$

Its "undisguised" inverse is then given by

$$\Sigma^{-1}(v) \begin{array}{ll} = -\alpha + (y_L + \alpha)\exp\left(\dfrac{v}{y_L + \alpha} - 1\right) & v \le \Sigma_L \\ = v - \alpha & \Sigma(y_L) \le v < \Sigma(y_R) \\ = y_R + \dfrac{\exp[K(v - y_R - \alpha) - 1]}{K} & \Sigma(y_R) \le v \end{array} \tag{12}$$

[2] De Guillaume, Rebonato, and Pogudin (2013) explain why they use the absolute-value operator instead of squaring the differences. We retain their choice.

Below we show how well, and how "universally," this function describes the level of volatility when negative rates are added to the data set. Before doing so, however, we must present the data used for the investigation. This we do in the following section.

THE DATA

For our study we used the USD government bond data available from Gurkaynak, Sack, and Wright (2007) for nominal rates,[3] and Gurkaynak, Sack, and Wright (2000) for TIPS.[4] These are the yields of "virtual" discount bonds obtained by least-square fit to the prices of nominal and real Treasuries using the popular Nelson-Siegel (1987) model. They cover the period December 31, 1971, to February 29, 2020.

For the GBP yields of virtual nominal and real discount bonds we made use of the data provided by the Bank of England.[5] As in the case of the USD data provided by the Fed, we arrived at the yields by using a variation of the Nelson-Siegel (1987) model. The period covered is January 31, 1970, to December 31, 2019 for nominal rates and December 31, 1984, to December 31, 2019, for real rates.

The EUR-DEM yields of virtual discount bonds were sourced from the Bundesbank,[6] and cover the period January 31, 1975, to December 31, 2018.

Finally, for the JPN data, which cover the period June 15, 1972, to June 15, 2007, we adapted data from the Japanese Ministry of Finance.[7]

The inflation data for USD used below to create a proxy for the inflation expectation were obtained from the CPI time series in the FRED database.[8] The data cover the period January 1947 to July 2020.

We stress that in the original work by De Guillaume, Rebonato, and Pogudin (2013) a variety of tenors and risk-less rates were used in order to substantiate the case for the universality of the relationship they uncovered. As we consider the universality point settled, we focus in this study on extending the empirical relationship to negative rates and on explaining the origin of this level dependence. Therefore, we do not analyze the 1-month, 3-month-, 6-month, and 12-month rates as De Guillaume, Rebonato, and Pogudin did, and we simply examine the yearly yields described above.

EMPIRICAL RESULTS

To the data described above we apply the procedure with the modified Sigma function. More precisely, first we fit, currency by currency, the best modified Sigma functions to the yields in USD, EUR, GBP, and JPN. When we do so, we obtain the disguised and undisguised yields in Exhibits 2 and 3, and the parameters reported in Exhibit 4. It is interesting to note how the individual calibrations produced very similar optimal parameters for the four currencies under study, with y_R ranging from 0.0835 (JPN) to 0.0600 (GBP), y_L ranging from 0.0145 (GBP) to 0.0334 (JPN), K between 25.2745 (JPN) to 33.4667 (EUR), and α between 0.0010 (USD) and 0.0220 (EUR). It is also noteworthy that the displacement coefficient for USD is non-zero, *despite no USD yields in our data set being negative*.

[3] available from http://www.federalreserve.gov/econresdata/researchdata/feds200805.xls.
[4] available from http://www.federalreserve.gov/pubs/feds/2008/200805/200805abs.html.
[5] available from https://www.bankofengland.co.uk/statistics/yield-curves.
[6] available from https://www.bundesbank.de/en/statistics/time-series-database.
[7] available from https://www.mof.go.jp/english/jgbs/reference/interest_rate/index.htm.
[8] available from https://fred.stlouisfed.org/tags/series?t=cpi.

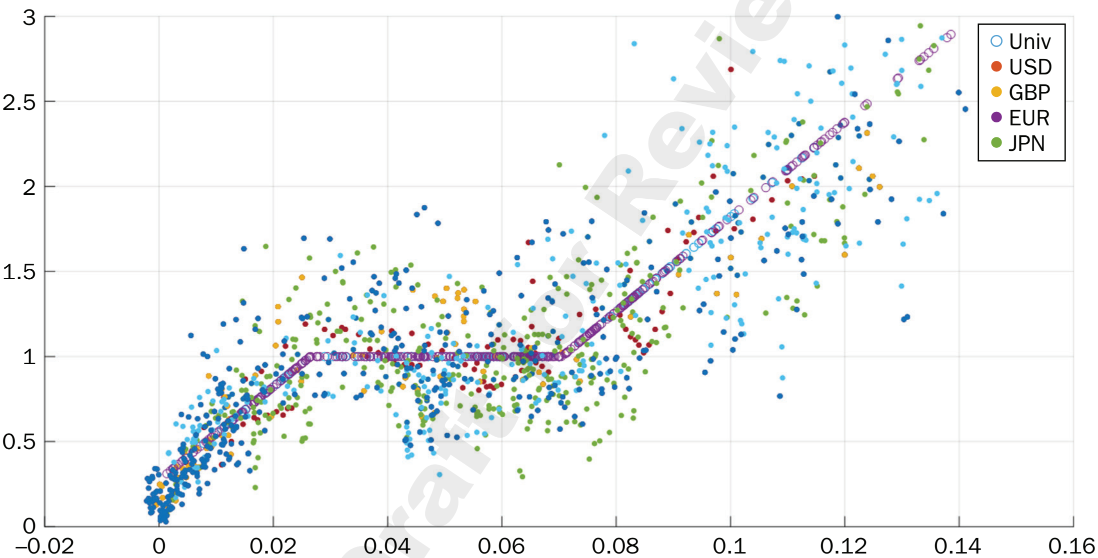
EXHIBIT 4

The Optimal Parameters y_L, y_R, K and the Displacement Coefficient, α, Determined Individually for the Four Currencies under Study (First Four Columns) and Using All the Data Combined

Currency	USD	EUR	GBP	JPN	Combined
y_L	0.0152	0.0165	0.0145	0.0334	0.0265
y_R	0.0719	0.0733	0.0600	0.0835	0.0705
K	32.3166	33.4667	26.3055	25.2745	27.8175
α	0.0010	0.0220	0.0177	0.0082	0.0103

EXHIBIT 5

The Rescaled Magnitude of the Movement in Rates as a Function of the Rate Level (in Percentage Points) for All the Currencies Combined, Obtained with the New Sigma Function and the Universal Parameters. Using 16 points in the averaging, and with the empty circles playing the role of the black line in Exhibit 1. See Appendix I and De Guillaume, Rebonato, and Pogudin (2013) for a precise description of the procedure



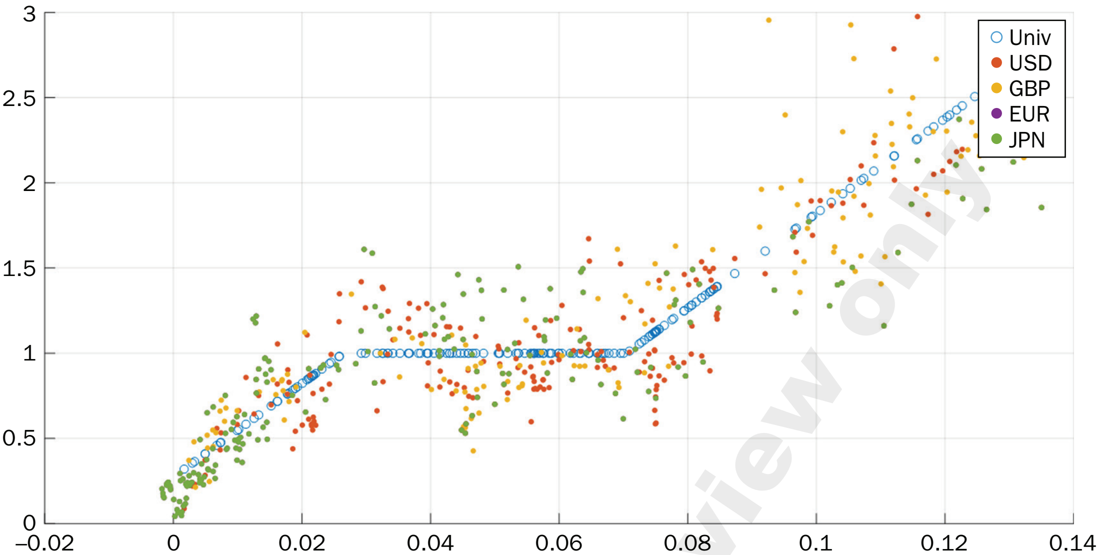
Next, we combine the data for the four currencies, and obtain the optimal parameters for the whole set. These are shown in the rightmost column of Exhibit 4. Not surprisingly, but reassuringly, the optimal parameters and the displacement coefficient obtained using all the data at the same time lie comfortably in the ranges for the individual currencies. Exhibits 4, 5, and 6 then show the rescaled magnitude of the movement in rates as a function of the rate level (in percentage points) for all the currencies combined, obtained with the new Sigma function and the same universal parameters using 16, 32, and 64 points in the averaging.[9] These are the equivalent of Exhibit 1 obtained with the new Sigma function. The qualitative similarity is apparent, as is the fact that the new function handles the negative rates very well.

Note that the variability around the new Sigma function (denoted by the empty circles) describes the residual degree of stochasticity of the volatility— i.e., the part of the variability of rates not captured by the level dependence of their volatility. We note first that a visual inspection of the figures suggests that this "volatility of volatility" is proportional to the level of rates. This impression is substantiated by a

[9] The averaging procedure is described in Appendix I, and the values of 16, 32, and 64 correspond to the parameter l discussed there.

EXHIBIT 6

Same as Exhibit 5, but Using 32 Points in the Averaging



regression of the absolute values of the differences between the scaled volatility and the Sigma function against the level of rates. The β coefficient of the regression (2.17) indeed indicates a positive dependence, and the associated *t*-statistic (8.54) suggests that the level variable is statistically strongly significant. (The R^2 of the regression is 0.210.)[10] We do not pursue this angle farther, but note that this empirical finding can be of use in the modeling of interest rates in a stochastic-volatility setting. The prevailing stochastic-volatility models, (such as the models in Hull-White 1987, Heston 1993, or Fouque, Papanicolau, and Sircar 2000, to name a few of the most popular ones), in fact, posit a level-independent volatility of volatility. However, to our knowledge, no justification, apart from simplicity and analytic tractability, has ever been given for this choice.

Going back to the main focus of our investigation, what lies at the root of the universal behavior that De Guillaume, Rebonato, and Pogudin first uncovered remains to be explained, and that, with some modifications, we confirm. We turn to this task in the next section.

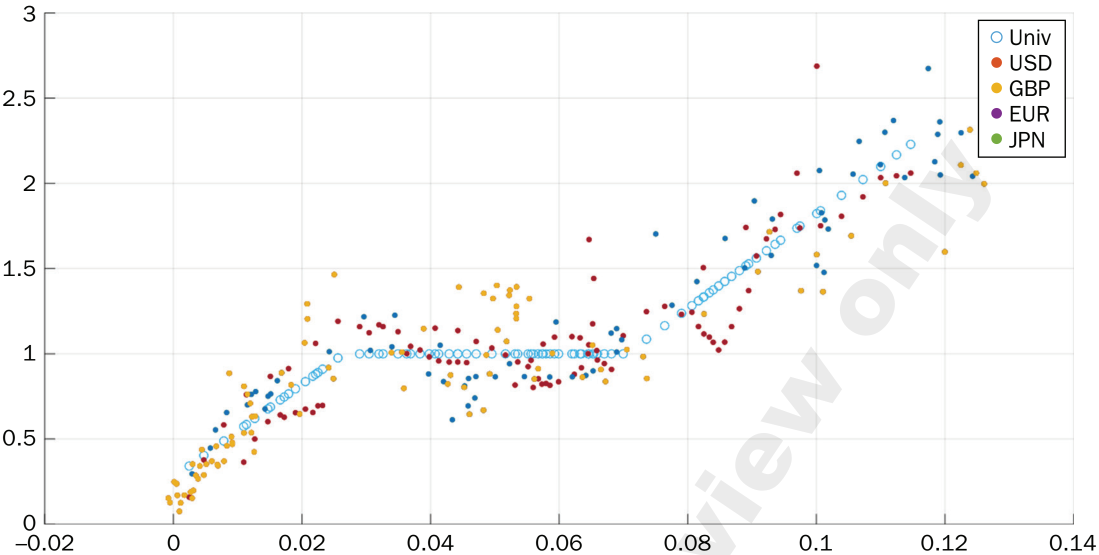
EXPLANATION OF THE RESULTS

Nominal rates can be thought of as the sum of real rates and inflation expectations (with the associated risk premiums, if any). We can ask whether the dependence of the volatility on the level of rates is inherited from a similar dependence of inflation volatility on the level of inflation, or of real-rate volatility on the level of real rates. To answer this question we apply the same standardization procedure we employed for nominal rates, first to inflation and then to real rates. We focus on USD and GBP, for which the message is clearest, and for which we can obtain real-rate information from traded market instruments (inflation-linked Treasury bonds).

[10] The results were obtained with an averaging parameter of 32, and were robust to different choices (16 and 64) for the number of points in the average.

EXHIBIT 7

Same as Exhibit 5, but using 64 Points in the Averaging



INFLATION ANALYSIS

Using the data described above we calculated the year-on-year inflation for USD. We then applied the standardization procedure, and obtained the results shown in Exhibit 8.[11] As is apparent, inflation volatility is at best mildly dependent on the level of inflation, and the resulting Sigma function is flat for all inflation levels. We note that there is an outlier with standardized volatility of almost five for extremely low values of inflation. If this point (which, however, cannot be an isolated outlier as it is the average of 16 individual points) is windsorized, there is a mild dependence of inflation volatility on its level, but the qualitative message does not change; the inflation component does not explain the steep increase in the volatility of nominal rates observed in high-yield environments.

REAL-RATE ANALYSIS

The picture changes radically when we look at real rates and their (standardized) volatility. In the case of GBP we have direct access to real yields thanks to the long history of prices for inflation-linked Treasury bonds (the UK "Linkers"), and this allows us the directly deploy our procedure to the time series of fitted real yields. We note that real rates have recently been very negative in GBP, and the use of the modified Sigma function is therefore essential for the analysis.

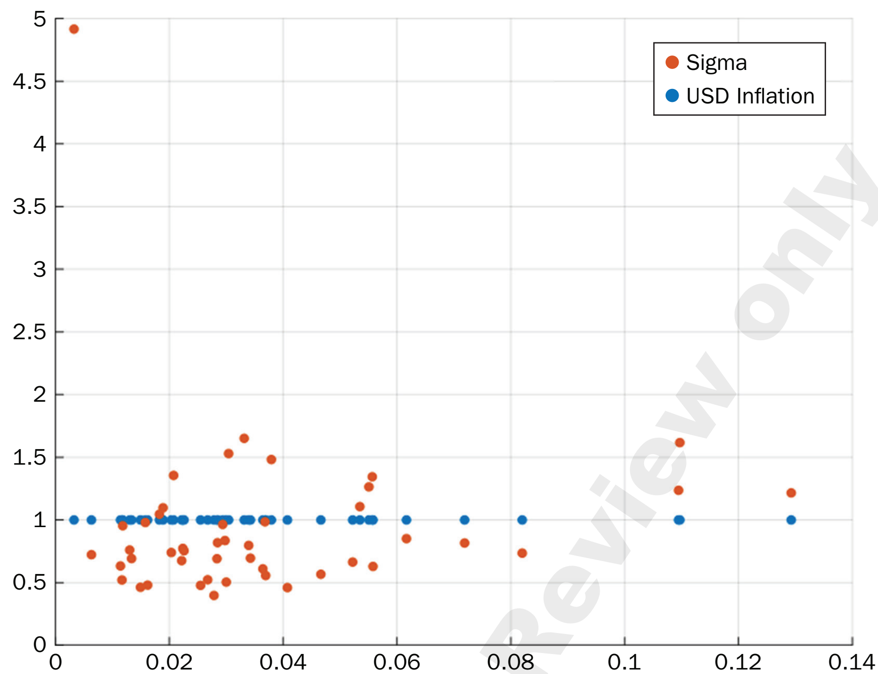
In the case of USD, we have a very short history for the inflation-linked TIPSs, and in this period US inflation has always been muted. We therefore proceed in a more indirect way. First, following the procedure discussed in detail in Cieslak and Povala (2015), we create an estimate of inflation expectations by calculating a slow-moving average of 10-year inflation (monthly realizations).[12] Given this proxy for inflation

[11] The figure was obtained with an averaging parameter of 16.

[12] We allow for the possibility that representativeness bias, as discussed, e.g., in Gennaioli and Shleifer (2018), may bias the estimate, by allowing for the latest realization of inflation to affect the expectation more strongly. When we do so, the qualitative results are unchanged, but we obtain a stronger relationship. For simplicity, we do not pursue this angle.

EXHIBIT 8

The Standardized Volatility of Inflation against the Inflation Level, Obtained Using 16 Points in the Averaging



expectations, we then estimate the real yield as the difference between the nominal rate and the inflation expectation proxy.

After obtaining these estimates of the real yields for GBP and USD, we use our procedure to obtain the dependence of the real-rate volatility on the level of real rates.

When we do so, we obtain the remarkable patterns shown in remains to be explained is remains to be explained is Exhibit 9. For high levels of real rates, two very distinct regimes clearly emerge, with a marked bifurcation in both currencies in the standardized volatility for levels of real rates above approximately 1% for GBP and 2% for USD. The pattern is striking, clear-cut, and remarkably similar in the two currencies for which we have reliable real-rate data, despite the different periods covered by GBP Linkers and USD TIPSs, and the different procedures employed to estimate the real rates. How can this complex behavior be reconciled with the much simpler pattern observed in the case of nominal rates? We answer this question immediately below.

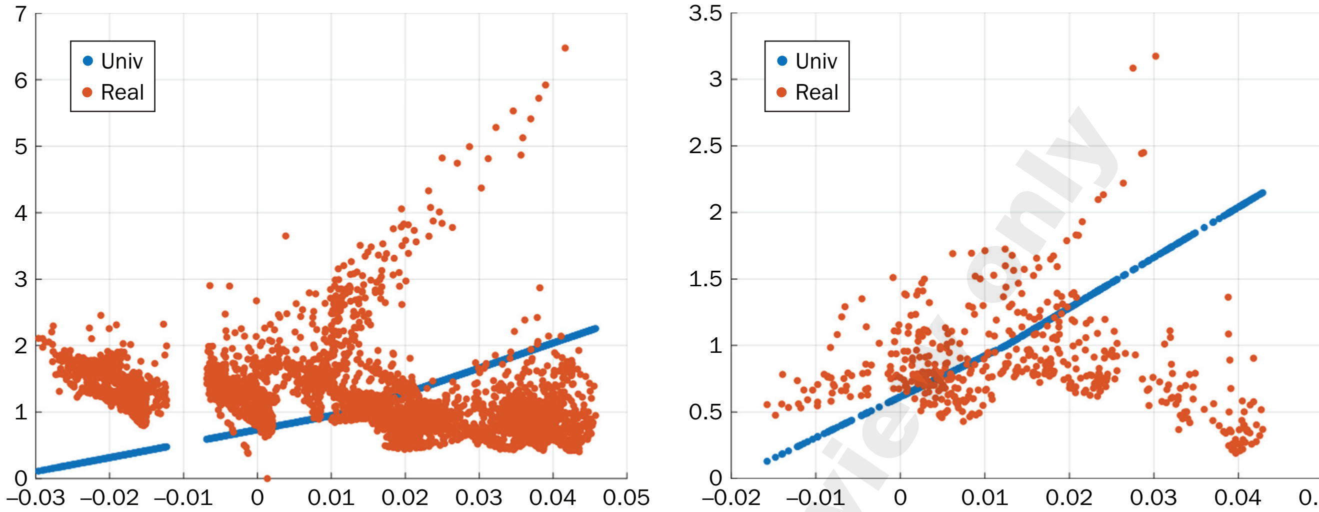
Combining the Inflation and Real-Rate Analysis

Naively superimposing the level dependence of the volatility found for inflation and real rates produces nothing resembling the three-section picture originally found by De Guillaume, Rebonato, and Pogudin, and, in its essence, recovered by this study (see Exhibits 5–7). The “bifurcated” pattern of the real-rate volatility is particularly puzzling. How can one make sense of these observations?

One possible explanation can be along the following lines. Suppose two distinct regimes of real-rate volatility exist when real rates are high: one (the lower volatility branch) linked to conditions of *low inflation*, and the other (the upper volatility branch) associated with *high inflation*. Since real rates rarely exceed the few percentage points, if our explanation were correct low inflation (expectations) plus high real rates would only produce “mid-sized” nominal rates. Since, as we have seen, inflation volatility barely depends on the level of inflation, the resulting nominal-rate volatility would

EXHIBIT 9

The Standardized Volatility of GBP (left panel) and USD (right panel) Real Rates against Their Levels, Obtained Using 16 Points in the Averaging



display little, if any, dependence on the level of *nominal* rates, as indeed we find in the central region of the three-section graph. However, when high real rates combine with high inflation (expectations), then the upper volatility branch of the real-rate volatility graph would be superimposed onto the level-independent volatility, giving rise to the ascending segment of the three-section graph for the nominal-rate volatility.

How can we test this hypothesis? If the explanation is correct, we should find a positive relationship between periods of high inflation and periods of high real rates. To test whether this is true we regress real rates in GBP and USD against inflation (expectations).[13] Since we do not have direct access to inflation expectations we proceed as follows. In the case of GBP, for which we have a long history of Treasury real bonds ("Linkers") covering periods of very high and very low inflation, we calculate the so-called average break-even inflation, i.e., the difference between the average 1-to-10-year nominal yields and the average 1-to-10-year real yields. We then regress the real yields against the break-even inflation, taken, modulo a risk premium, as a good proxy for the inflation expectations. In the case of USD, we make use of the inflation estimates obtained, and calculate the real rates as the difference between nominal rates and inflation expectations.

After obtaining these different estimates for real rates and inflation (expectations), we carry out the regressions

$$avg_t^{infl,x} = \alpha + \beta avg_t^{real,x} + \epsilon_t \tag{13}$$

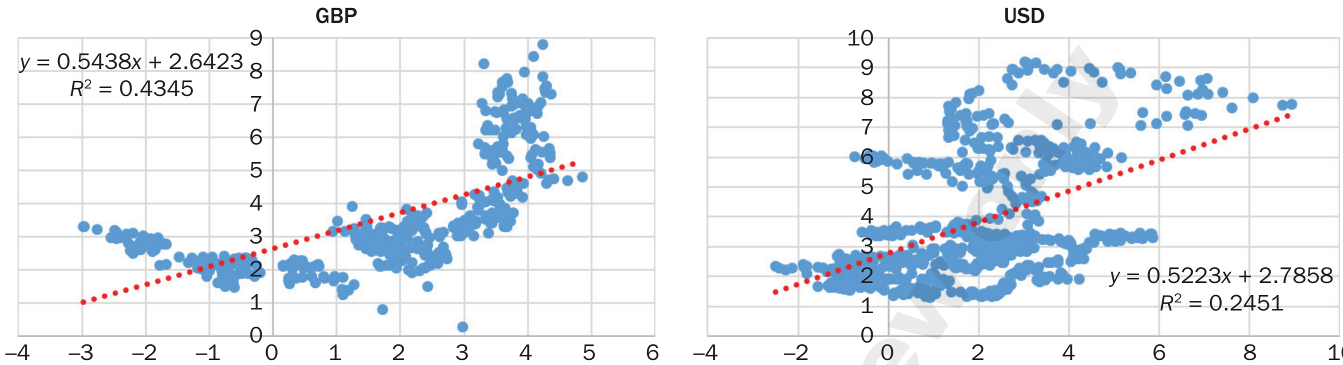
(In Equation 13 the symbol x denotes USD or GBP.)

When we do so, we find a strongly significant statistical relationship between the level of inflation and the level of real rates, with a very high t-statistic for the slope (14.62 for USD and 17.96 for GBP), positive, and extremely similar, β coefficients in both currencies (0.543 for GBP and 0.522 for USD), and a strongly significant R^2 coefficient (0.43 for GBP and 0.24 for USD). Exhibit 10 displays the scatter plots

[13] Strictly speaking, risk premiums should be added both to the real-rate and inflation contribution to nominal rates in both currencies. The argument does not change as long as we consider both components together with their risk premium.

EXHIBIT 10

The Regression of the Expected Inflation (*y* axis) against the Average Real Yield (*x* axis) for GBP (left panel) and USD (right panel). The inflation expectations were proxied by the break-even inflation in the case of GBP, and as the difference between the average nominal yield and the inflation expectation built as discussed in the text. Inflation expectations and real yields are in percentage points



of the average real yields and the inflation expectation proxies, clearly showing that the highest real rate realizations are strongly associated with the highest levels of inflation expectations.

In the case of GBP, where real rates have attained very negative levels, the left panel of Exhibit 10 shows that the nature of the relationship between break-even inflation and real yields is clearly more complex than linear, especially in the very-low-real-rate region, and suggests a parabolic behavior, which would warrant an independent analysis. For the purpose of our discussion, however, the link between high real rates and high inflation expectations—what matters for our explanation to hold—is clearly valid. If anything, if we carried out the regression to the top half of the real rates (the area where the bifurcation that we are trying to explain occurs) the linear relationship in Equation 13 would be even stronger (with an R^2 above 0.50).

These regressions therefore indicate that high real rates are indeed associated with periods of high inflation. So, when inflation is low, the level-independent lower branch of the bifurcation applies even if real rates are high. In this case the resulting nominal yields are of medium size, and we are in the central (level-independent) section of the "universal plot." When inflation is high, on the other hand, real rates tend to be high, the upper branch applies, and the associated nominal yields are also high. This gives rise to the rising right-most part of the "universal plot."

CONCLUSIONS

We have presented a straightforward extension (based on replacing a geometric diffusion with a displaced diffusion when rates are low) of the "universal relationship uncovered by De Guillaume, Rebonato, and Pogudin (2013) to handle the case of negative rates. After the same standardization they proposed, we confirm that the identical three-segment currency-independent ("universal") relationship exists in all currencies under study for the dependence of rate volatilities on their levels.

We have also observe that, again in all currencies, the volatility of volatility displays a clear positive dependence on the level of rates, and hence of the volatility itself. This can be of help in more realistic modeling of the stochastic volatility of interest rates.

Our more interesting contribution, however, is toward understanding *why* the three-segment relationship exists in the first place. We have shown that, of the two constituents of nominal rates, the volatility of inflation has a very mild dependence on the level of inflation. The volatility of real rates, however, shows a clear and remarkable "bifurcation" in the high-real-rate regime. By finding a strong association between periods of high real rates and periods of high volatility, we offer a simple and convincing, if somewhat "mechanical," explanation of the "universal relationship."

Several interesting, and, arguably more fundamental, questions remain unanswered. Why, for instance, are real rates more volatile when inflation is high? Why are real rates and inflation so strongly positively correlated? Why does the behavior of GBP inflation display a quasi-parabolic behavior as a function of real rates in the low-yield area? What are the common financial mechanisms at play in GBP and USD to produce the almost identical bifurcated pattern of Exhibit 9? We do not have answers to these questions, but we hope that the findings presented in this work can lay the foundations for finding them.

APPENDIX I

THE PROCEDURE TO CREATE EXHIBIT 1

The procedure to create Exhibit 1 is as follows.

- Choose the optimum Σ function using the methodology in our study.
- For each time series i create the lists $\overline{Y}_i(1), \ldots, \overline{Y}_i(m_i), \Delta_i(1), \ldots, \Delta_i(m_i)$ and calculate σ'_{Gi} and μ_i as above.
- For each i simultaneously re-order the sequences $[\Delta_i(j)]_{j \le mi}$ and $[\overline{Y}_i(j)]_{j \le m_i}$ so that the latter sequence is in ascending order.
- For each i split the the sequences $[\Delta_i(j)]_{j \le mi}$ and $[\overline{Y}_i(j)]_{j \le m_i}$ into equal sections of length l.
- For each i and for the kth section define $\overline{\mu}_{ki}$ to be the average of the kth converted yield section. Define $\mu_{ki} = \Sigma^{-1}(\overline{\mu}_{ki})$.
- For each i and for the kth section define $\zeta^*_{k_i}$ to be the average deviation of the kth Δ section. Define $\zeta_{k_i} = \zeta^*_{k_i} / \sigma'_{Gi}$, where the σ function σ_i is defined using the parameters y_L, y_R, and K from the optimized function and $\sigma_G = 1$. (The division by σ'_{Gi} ensures that the time series are compared to on a like-for-like basis).
- Plot the quantities ζ_{ki} against the average of the μ_{ki} sections, using a different color for each time series.

To produce the black line in Exhibit 1, one can plot the σ function corresponding to Σ, with σ_G set equal to 1. The quantity l—the number of points in the various averages—determines the degree of resolution, and is called in the body of the paper the "averaging parameter."

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